

Assessing the Strengthening Effect of Various Near-Surface-Mounted FRP Reinforcements on Concrete Bridge Slab Overhangs

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Abstract: The near-surface-mounted (NSM) method has proved to be a reliable alternative to the existing externally bonded (EB) method for the repair and strengthening of concrete structures using fiber-reinforced polymer (FRP) composites. This technique is especially advantageous in bridge deck upgrades for larger barriers and slab overhang strengthening. This paper presents the results of a comparison of the flexural behavior of bridge slab overhangs strengthened in negative bending moment regions with various types of NSM reinforcement that differ in surface condition (e.g., textured and sand coated), cross-sectional shape (e.g., round and square), material type (carbon and glass), and prestressing effect. Eleven full-scale overhang specimens (1.524 m long in overhang and 0.914 m wide) were tested under a cantilever condition. Test results showed that the FRP NSM reinforcements were effective in increasing both yield and ultimate strength of predamaged slab overhangs. All surface treatments were more beneficial than the smooth condition, and the square-shaped reinforcement displayed better performance than the round shape. The prestressing unit developed in this study is simple to apply and could be further explored for field applications. DOI: [10.1061/\(ASCE\)CC.1943-5614.0000182](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000182). © 2011 American Society of Civil Engineers.

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Introduction

Recent bridge inventory data show that many existing reinforced concrete (RC) bridges are experiencing slab deterioration with significant capacity loss in the superstructure [Federal Highway Administration (FHWA) 2009]. Meanwhile, structural upgrades are constantly demanded by increasing public transportation to accommodate greater live loads, larger barriers, and wider traffic lanes. Quickly and effectively repairing cracks and upgrading these superstructures with minimum traffic disruption has always been a challenge. A recent repair and strengthening technique based on fiber reinforced polymer (FRP), the near-surface-mounted reinforcement (NSMR) method, has shown unique advantages over many conventional methods, such as external cable strengthening, addition of steel rebar, and even the same FRP-based method using externally bonded reinforcement (EBR). Instead of external attachment, the NSM FRP reinforcement is embedded in the precut sawn grooves or slots in the concrete cover of the structural member, thus alleviating the premature debonding and durability issues commonly encountered in the EBR method. In addition, the NSM technique has minimum disturbance to the original aesthetics of the structure and has proved to be more efficient than the EBR

technique with much less brittle failure modes (e.g., Barros and Dias 2006; Hassan and Rizkalla 2003; El-Hacha and Rizkalla 2004).

Remarkable progress has been made in the existing literature to understand the behavior of RC members strengthened or retrofitted with NSMR, such as carbon FRP (CFRP) and glass FRP (GFRP). The main effort has been devoted to the areas including but not limited to (1) comparison of the NSMR and EBR methods, such as the flexural strengthening of T-shaped and rectangular RC beams using GFRP and CFRP NSM strips (El-Hacha and Rizkalla 2004; Quattlebaum et al. 2005) and CFRP NSM rods (Jung et al. 2005), shear strengthening of RC beams using CFRP NSM strips (Barros et al. 2006; Dias and Barros 2005; Bianco et al. 2010); (2) investigation of bond characteristics including effects caused by groove size (Kang et al. 2005), groove depth and spacing (Hassan and Rizkalla 2003; De Lorenzis and Nanni 2001), addition of grooves in the perpendicular direction for mechanical interlock (Jung et al. 2005), embedment length (Novidis and Pantazopoulou 2005; Hassan and Rizkalla 2003), local bond stress-slip relationship (Sena Cruz and Barros 2004; Sena Cruz et al. 2006; De Lorenzis and Nanni 2002; Achillides and Pilakoutas 2004), and adhesive type (Novidis and Pantazopoulou 2005; Shield et al. 2005; Teng et al. 2006); (3) NSM reinforcement type and amount (El-Hacha et al. 2005; Kishi et al. 2005), reinforcement size (Tang et al. 2006), orientation (De Lorenzis et al. 2000; Barros et al. 2006; Shield et al. 2005), and deformed versus sandblasted surface condition in terms of bond behavior (De Lorenzis and Nanni 2002); (4) use of prestressing, which is found to result in smaller and more uniformly distributed cracks and increased durability and stiffness of the structure (Nordin and Taljsten 2006; Triantafillou et al. 1992), although drawbacks exist, such as reduced strain capacity of the FRP materials and strain loss at the ends of NSMR, coupled with the end-peeling failures (El-Hacha et al. 2003; Wight et al. 2001); and (5) fatigue behavior of NSM-strengthened RC

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structures, such as the 40-year-old C-channel prestressed concrete girders using CFRP NSM bars and strips (Rosenboom and Rizkalla 2005) and medium-scale beams with NSMR (Quattlebaum et al. 2005; Aidoo et al. 2006).

The existing FRP NSM reinforcements are commercially available in various cross-sectional shapes (round, square, and strip), surface configurations (e.g., smooth, sand coated, spirally wound, ribbed, and textured) and material types (e.g., glass, carbon, and aramid). The current literature provides partial evaluation of some of these factors and mostly evaluate the concern for bond strength at the interface [e.g., De Lorenzis and Nanni 2002; De Lorenzis et al. 2004; Achillides and Pilakoutas 2004; American Concrete Institute (ACI) 2004, 2008; Baena et al. 2009]. Further research is needed to evaluate the structural-level performance of the deficient concrete bridge superstructures strengthened with the aforementioned different types of NSMR. In addition, previous studies have primarily been focused on RC beams and decks in their positive bending moment regions, with limited research on concrete slabs for which NSMR is applied in the negative bending moment area, such as bridge slab overhangs. Teng et al. (2000) studied the flexural behavior of strengthened cantilever slabs, but the externally bonded GFRP strips were investigated instead of the NSM reinforcement. Al-Mahmoud et al. (2010) conducted tests using NSM CFRP rods in reinforced members, but the study was focused on cantilever concrete beams instead of slabs. Moreover, as pointed out in that paper, cantilever beams strengthened with FRP have a specific problem called *strengthening outward pressure* (unlike the simply supported cases in which the pressure is restrained by the two supporting ends). Although their test results on three selected beams showed that cantilever beams strengthened by CFRP rods exhibited similar failure modes to the classical beams in four-point flexure, the validity of this conclusion in slab strengthening cases is still questionable. Also, the existing direct prestressing methods of NSMR members require complex anchorage systems that are very labor-intensive and are practically difficult for many on-site bridge strengthening projects. Therefore, this research attempts to address these practical issues to allow for more applications of NSMR in concrete bridge slab overhang strengthening and rehabilitation. In this study, an experimental program was conducted in which 11 full-scale RC slab overhang specimens (1.524 m long in overhang and 0.914 m wide) were tested under a cantilever condition with the load applied at the free end. To simulate the field condition of an existing bridge, the overhang slabs were precracked (predamaged) during the test before the application of NSMR on the top surface of the slabs for flexural strengthening. This precracking effect is considered to be of significant practical interest (De Lorenzis and Teng 2007) because it affects crack propagation and the debonding failure process of strengthened members (a direct comparison with uncracked cases is beyond the scope of this current study). Test results

showed that the FRP NSM reinforcements effectively increased both the yield and the ultimate loads of predamaged slab overhangs. All surface treatments were more beneficial than the smooth condition, and the square-shaped reinforcement displayed better performance than the round-shaped reinforcements. The prestressing unit developed in this study is simple to apply and could be further explored for field applications.

Experimental Program

Materials

Two separate batches of slab specimens were constructed because of space constraints inside the laboratory. The average 28-day concrete compressive strength for the first and second batch of specimens was obtained as 32.4 and 36.7 MPa, respectively, following the testing method per ASTM C39/C39M (ASTM 2005). The specimens were designed as conventional RC overhang slabs consistent with the California Department of Transportation (Caltrans) bridge design requirements (Caltrans 2008) and the LRFD specifications (AASHTO 2007). No. 16 steel bars with a nominal diameter of 15.875 mm were used for the main steel reinforcement in the slab, and No. 13 rebars with a nominal diameter of 12.7 mm were placed for shrinkage and temperature control. Because the flexural strength of the specimens could potentially exceed their shear strength due to NSMR strengthening, shear stirrups made of No. 10 steel bars (with a nominal diameter of 9.525 mm) were added to all testing specimens to prevent such undesired shear failure in the test. The yield strength of the No. 16, 13, and 10 steel reinforcing bars was 493, 476, and 473 MPa, respectively, with the corresponding tensile strength of 740, 745, and 690 MPa, respectively, per manufacturer's data.

Eight types of FRP NSM reinforcement were investigated in this study (Fig. 1): sand-coated-only (SC) CFRP round rod, ribbed CFRP round rod, textured CFRP strip (side surfaces have a roughened texture), spirally wound plus sand-coated (SWSC) CFRP round rod, smooth CFRP square rod, spirally wound plus sand-coated GFRP round rod, smooth CFRP round rod, and textured CFRP round rod (surface has a roughened texture). The material properties of all FRP NSM reinforcements are summarized in Table 1 based on the manufacturer's data. A typical two-part epoxy resin was used for the adhesive, which has a tensile strength of 72.4 MPa.

Specimens

A total of 11 concrete slab overhang specimens were constructed. The first batch included specimens S1, S4, and S5, and the second batch had the rest (S6–S13). Each specimen had an overhang length of 1.524 m and a width of 0.914 m, as illustrated in Fig. 2. The thickness of the slab varied continuously from 305–229 mm

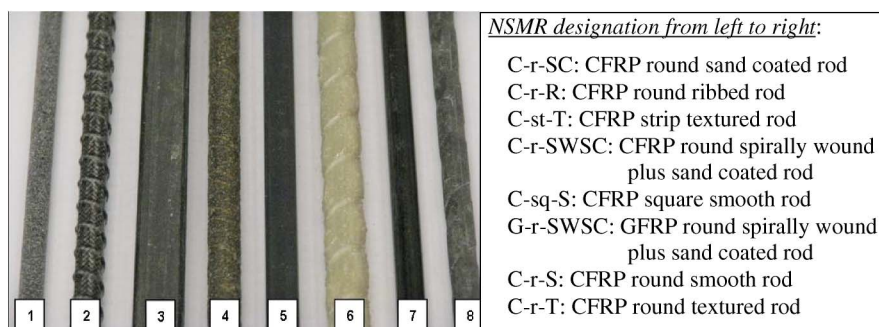


Fig. 1. Illustration of various FRP NSM reinforcements

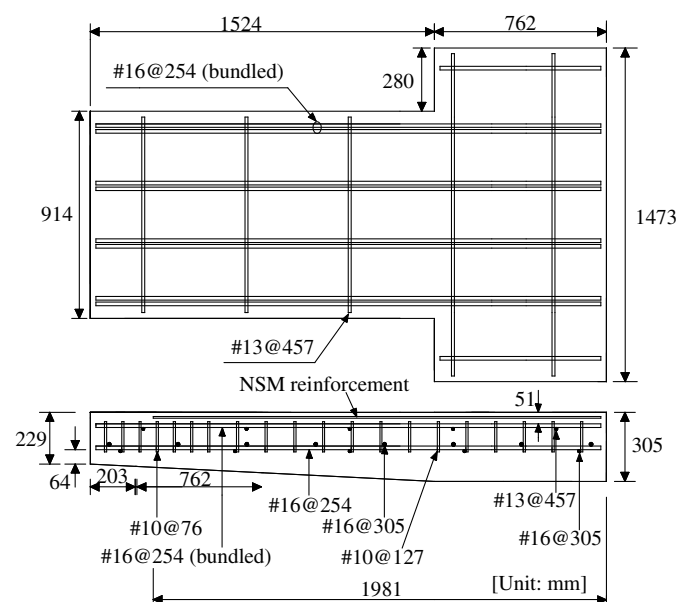
Table 1. Material Properties of FRP NSM Reinforcements

NSMR designation ^a	FRP type	Sectional shape	Size ^b (mm)	Sectional area (mm ²)	Perimeter-to-area ratio	Surface treatment ^c	Tensile strength (MPa)	Elastic modulus (GPa)	Tensile strain (%)
C-r-SWSC (4)	Carbon	Round	9.5	70.96	0.42	SWSC	2,340	142.9	1.64
G-r-SWSC (6)	Glass	Round	12.7	129.00	0.31	SWSC	690	40.8	1.60
C-r-S (7)	Carbon	Round	9.5	70.96	0.42	S	2,800	155	1.80
C-sq-S (5)	Carbon	Square	10	100.00	0.40	S	2,000	155	1.50
C-r-SC (1)	Carbon	Round	7.5	44.00	0.53	SC	2,300	130	1.80
C-r-R (2)	Carbon	Round	9	65.20	0.44	R	2,185	150	1.43
C-r-T (8)	Carbon	Round	9	65.20	0.44	T	2,068	124	1.70
C-st-T (3)	Carbon	Strip	4.5 × 16	71.30	0.57	T	1,965	124	1.50

^aFollowing a labeling scheme “material-shape-surface (picture number in Fig. 1),” for example, C-st-T (3) represents CFRP strip-shaped textured NSMR as marked by #3 in Fig. 1.

^bDiameter for round rods; edge length for square rods; thickness by width for rectangular strips.

^cSC = sand-coated-only; R = ribbed; T = textured; SWSC = spirally wound sand-coated; S = smooth.

**Fig. 2.** Overhang specimen dimensions and reinforcement details

from the clamped end to the free cantilever tip to allow for a tapered cross-sectional design in the slab (Caltrans 2008). Specimens S1 and S6 were unstrengthened RC slabs serving as the control specimens, and the rest were strengthened with eight different types of NSM rods or strips on the top after experiencing precrack (see summary in Table 2). Following the ACI 440.2R-08 guide (ACI 2008), the total axial stiffness ($E_f A_f$) of the various types of NSM reinforcement, as commonly used to compare the flexural behavior of strengthened concrete beams (e.g., El-Hacha and Rizkalla 2004; Kishi et al. 2005), was designed to be the same among all strengthened specimens to allow for direct comparisons of all specimens. E_f is the Young's modulus, and A_f is the total cross-sectional area of the FRP. To model the cantilevered end condition in bridge slab overhangs in the field (continuity condition at the support), a wider slab section was designed at the supporting end of each specimen. The length of the section was determined based on the required development length of NSM reinforcement per ACI 440-2R.08 guidelines, which was calculated as 724 mm on average for a 1.524 m long slab section tested in this study.

Test Setup and NSM Application

Some existing research shows that a concrete cover separation failure could occur when the total embedment length of NSMR is

Table 2. Test Parameters

Specimen designation ^a	NSM reinforcement type ^b											
	Material		Cross-sectional shape			Surface treatment					Prestress	
	C	G	r	sq	st	S	SWSC	SC	R	T	Yes	No
S1												
S4C-r-SWSC	3 ^c		•				•					•
S5G-r-SWSC		5	•				•					•
S6												
S7C-r-S	3		•			•						•
S8C-sq-S	2			•		•						•
S9C-r-SC	5		•					•				•
S10C-r-R	3		•						•			•
S11C-r-T	3		•							•		•
S12C-st-T	3				•					•		•
S13C-st-T-p/s	3				•					•	•	

^aFor NSMR strengthened specimens, following a labeling scheme “specimen#Material-shape-surface,” for example, S12C-st-T represents Specimen#12 with CFRP strip-shaped textured NSMR (no prestress, Specimen#S13 is the only one applied with prestressing, i.e., S13C-st-T-p/s).

^bC = CFRP, G = GFRP; r = round, sq = square, st = strip; S = smooth; SWSC = spirally wound plus sand-coated; SC = sand-coated-only; R = ribbed; T = textured.

^cTotal number of NSM rods or strips used in the specimen.

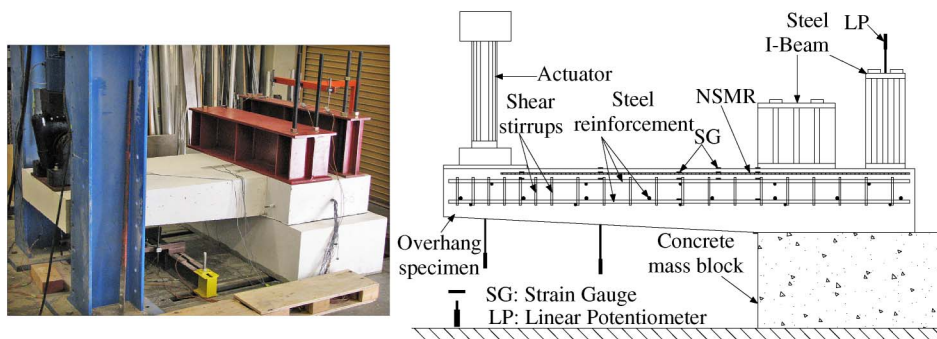


Fig. 3. Test setup configuration

insufficient [e.g., Teng et al. (2006) reported a shifting of failure mode from concrete cover separation to concrete crushing as the embedment length increased]. In reality, this cover separation failure should be avoided in overhang strengthening by embedding the NSM rods or strips over an extended length into the inner span of the bridge to leave a sufficient total embedment length of NSMR. Therefore, to preclude this undesired failure mode and to allow for direct comparisons on various NSM reinforcements in this study, each specimen was clamped down onto a heavily reinforced concrete mass block by using steel I-beams at the supporting end. The entire unit was then tied down to the lab strong floor through high tension bolts. The load was applied at the free end by using a closed-loop servocontrol hydraulic actuator with a capacity of 1,000 kN under a displacement control at a speed of 1.143 mm/min. Fig. 3 illustrates the overall configuration on the test setup. A series of typical foil strain gauges (SG) were attached to the concrete top and bottom surfaces (with a gauge length of 30 mm), steel rebars (6 mm gauge length), and NSM bars or strips (5 mm gauge length) along the specimen length. Spring-based linear potentiometers (LPs) were installed under the specimen at midspan and free end (directly under the loading point) to measure the vertical deflection. The steel I-beams used at the clamped end were also monitored with LPs for any uplifting during the test. Load, vertical deflection, and strain responses were recorded using a National Instruments data acquisition system in the lab.

The control specimens (S1 and S6) were tested monotonically up to failure to determine the flexural capacity of the designed normal RC slab. The other nine specimens were precracked first under a load of 35.6 kN to introduce a visible amount of damage (cracks) in the slabs. This load level was determined on the basis of the observations from the unstrengthened specimens (S1 and S6), in which the initial hairline crack was developed at a load level of 22.2 kN, but obvious damage was not observed until the load reached 35.6 kN. The specimens were then removed from the testing rig to apply the NSM strengthening. The top surface of the precracked specimens was cut by a concrete saw with a guide rail to ensure straight grooves. Depending on the size and type of the NSM reinforcement, four different groove sizes were selected [per ACI 440.2R-08 and existing literature, such as De Lorenzis and Nanni (2001) and Sena and Barros (2004)]: (1) 19-mm wide and deep for the smooth CFRP square rods, SWSC CFRP round rods, smooth CFRP round rods, ribbed CFRP round rods, and textured CFRP round rods; (2) 25.4-mm wide and deep for the SWSC GFRP round rods; (3) 17.5-mm wide and deep for the SC CFRP round rods; and (4) 17.5-mm wide and 25.4-mm deep for the textured CFRP strips. After the surfaces of the grooves were cleaned and dried, epoxy adhesive was poured into the grooves to fill up to approximately half of the depth of the grooves. NSM bars or strips

were then carefully placed into the center of the grooves, and additional adhesive was applied on the top to completely fill the grooves. The epoxy adhesive was allowed to cure at room temperature for at least 7 days before the strengthened specimen was moved back and tested up to failure (with several short pauses to allow for crack inspection).

Prestressing Design

To improve the material efficacy of FRP reinforcement, direct prestressing of NSMR members has been studied in the literature, but its application in the field is limited because of practical constraints, such as accessibility to the site and intensive labor work involved. The prestressing unit specially developed in this study aims to provide a fast, efficient, and economical prestressing method for feasible applications in the field. For illustration purposes, the prestressing unit designed here was only applied to the CFRP strips (Specimen S13C-st-T-p/s). It consists of two tubular steel sleeves applied outside the two ends of the NSM strip (filled with epoxy adhesive in the gap) and two anchorage rings at the far ends. At one end, the steel sleeve is welded to the anchorage ring cut from a large steel tube to form an eye-bolt shape, as shown on the right end in Fig. 4(a). At the other end, the steel sleeve is welded to a threaded rod, which is then connected to the anchorage ring by using a nut and washer [the left end in Fig. 4(b)]. To prevent the sleeve from rotating during prestressing, a small steel disk is welded to the outside of the sleeve inside the groove [Fig. 4(a)]. To host the flat anchorage rings, two round holes are predrilled on the top surface of concrete in addition to the grooves using a concrete core drill. After the entire unit is positioned in the middle of the slot, a prestressing force is applied by tightening the nut using a wrench [Fig. 4(c)]. This prestressing force corresponds to approximately 25% of the ultimate strain in the NSM strip by monitoring the strain gauge readings in the strip during the prestressing operation. Because of losses, this prestressing level reduced to 17.5% before the test. The gap inside the groove is then filled with epoxy adhesive for the final finishing.

Test Results and Discussion

Effect of Material Type

The control specimen (S1) showed a typical flexural failure in which the bottom concrete crushed at the clamped edge with the steel reinforcement already yielded at the top layer. A load-carrying capacity of 170 kN was reached at failure with a maximum tip deflection of 64.4 mm and a maximum crack width of 4 mm. For Specimen S4C-r-SWSC strengthened with 3 SWSC CFRP rods, a small cracking sound was noticed in the epoxy cover as the load

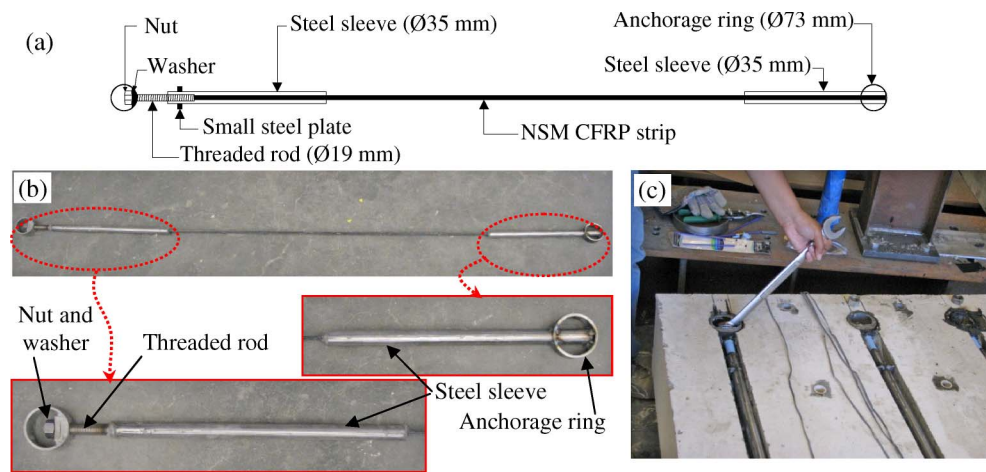


Fig. 4. Design of developed prestressing unit

exceeded 157 kN. Final failure occurred when the load reached 214 kN, and the epoxy cover on the CFRP bars was broken and concrete crushed at the bottom of the slab. The CFRP rods in S4C-r-SWSC ruptured at failure, where the tensile strain was measured as $16,432 \mu\epsilon$ (reaching their ultimate strain level of 1.64%, as shown in Table 1). Similar crack propagation, failure mode, and load-carrying capacity were observed in Specimen S5G-r-SWSC, which was strengthened with five SWSC GFRP rods. In comparison to the unstrengthened slab (S1), the load-carrying capacity of the strengthened slabs was increased by 25.9% and 28.2% for S4C-r-SWSC and S5G-r-SWSC, respectively. The load level at steel yielding was also increased by 24.6% and 29.8% for S4C-r-SWSC and S5G-r-SWSC, respectively, as shown in Table 3 and Fig. 5. The maximum strain in the GFRP bars in S5G-r-SWSC was measured to be $14,002 \mu\epsilon$, which was about 88% of its ultimate strain limit. These results indicate that with the same amount of axial stiffness, the SWSC GFRP and CFRP round rods can provide the equivalent strengthening effect for concrete overhanging slabs. Similar observation is also reported in an earlier study on reinforced concrete beams strengthened with NSM strips that are made of CFRP and GFRP (El-Hacha and Rizkalla 2004).

Effect of Cross-Sectional Shape

Round, square, and strip (rectangular) shapes of NSMR are included in this study. With an equivalent level of axial stiffness, Specimen S7C-r-S contained three smooth CFRP round rods, and S8C-sq-S had two smooth CFRP square rods. Specimen S11C-r-T was strengthened with three textured CFRP round rods, and Specimen S12C-st-T had three textured CFRP strips. The control specimen during this batch (S6) behaved very similarly to the control specimen in the previous batch (S1). The failure of S7C-r-S and S8C-sq-S originated in localized splitting of epoxy adhesive in the grooves, which were penetrated by the extensive transverse cracks in concrete due to flexure. The final failure was caused by concrete compressive failure (crushing) at the slab bottom toward the support. Both of the strengthened specimens had greater yield strength than the control specimen, i.e., 15.4% for S7C-r-S and 24.5% for S8C-sq-S, as shown in Table 3 and Fig. 6. The strains in the FRP rods at failure were found to be 6,250 and $7,840 \mu\epsilon$ for S7C-r-S and S8C-sq-S, respectively, which are both less than 60% of their ultimate strain limits. Specimens S11C-r-T and S12C-st-T both failed because of compressive failure in concrete and splitting of epoxy adhesive in the grooves. No rupture of the textured CFRP rods or strips was observed during the test.

Specimen S11C-r-T (round rods) had a yield load 6.6% greater and a failure load 31.8% greater than the control specimen (S6). In comparison with S11C-r-T (rods), Specimen S12C-st-T (strips) showed a failure load close to S11C-r-T, but it had a greater yield load increase with respect to S6 (18.9%), as shown in Fig. 6. A maximum tensile strain of 15,100 and $14,300 \mu\epsilon$ was measured at failure in the round rods (S11C-r-T) and strips (S12C-st-T), respectively, which was about 88.8% and 95.3% of their corresponding strain limits (Table 1).

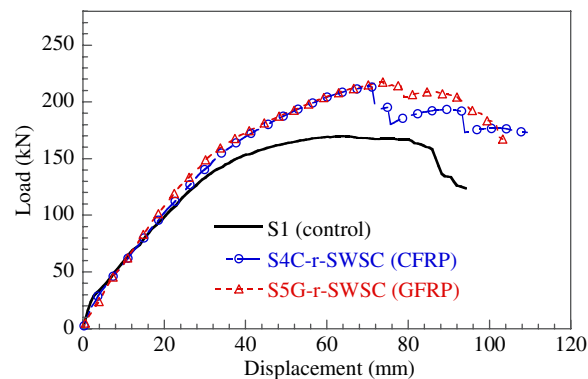
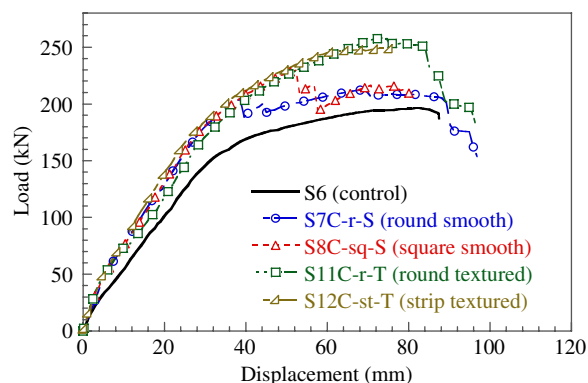
Based on these test results, the smooth CFRP round and square rods both contributed to the load increase in the slab at yielding and ultimate levels. The perimeter-to-sectional area ratio of the round and square rods are fairly close (0.42 for the round rod, and 0.4 for the square rod). During the test, Specimen S8C-sq-S reached a greater load at yielding level than S7C-r-S (slightly greater at the ultimate level). This might be attributed to the wedging effect (Achillides and Pilakoutas 2004) caused by the sharp edges of the square cross section in S8C-sq-S that improved the bond of the rod and consequently helped on the load-carrying capacity. This wedging effect was reported in the direct pullout tests on GFRP and CFRP bars (Achillides and Pilakoutas 2004), in which square bars showed a 25% greater bond strength than round bars because the wedging effect is more important in the case of square bars because of their sharp edges, according to the authors. The specimens strengthened with CFRP round rods (S11C-r-T) and strips (S12C-st-T) also had greater loads than the control slab, with a similar ductility between them but a greater yield load in S12C-st-T than S11C-r-T. Given equivalent axial stiffness of NSMR, the strip shape could be more efficient than the round shape because the strip has a larger perimeter-to-sectional area ratio than the round shape, which could result in lower bond stresses along the NSMR surface (in this case, the perimeter-to-sectional area ratio is 0.44 for the round rod and 0.57 for the strip).

Effect of Surface Treatment

Five existing surface configurations are investigated for CFRP round rods: spirally wound plus sand-coated (S4C-r-SWSC), smooth (S7C-r-S), sand-coated-only (S9C-r-SC), ribbed (S10C-r-R), and textured (S11C-r-T). Test results showed that all specimens failed because of crushing in concrete in the compression side together with some epoxy splitting failure in the grooves. The maximum strain in the CFRP rods was found to be all close to their corresponding ultimate limits, except for the smooth rods in S7C-r-S (Table 3). Fig. 7 compares the load versus displacement

Table 3. Summary of Deflection, Load, Curvature, Strain, Ductility, and Failure Modes

Specimen designation	Initial cracking ^a			Yielding			Ultimate			Ductility index		Failure mode ^d	
	δ_{cr} (mm)	P_{cr} (kN)	δ_y (mm)	P_y (kN)	$\Delta P/P^b$ (%)	χ_y (m ⁻¹)	δ_u (mm)	P_u (kN)	$\Delta P/P$ (%)	χ_u (m ⁻¹)	ε_{frp} (%)	RC slab	NSM FRP
S1	1.86	22.2	27.8	126	— ^c	0.015	64.4	170	—	0.031	—	f + cy	—
S4C-r-SWSC			35.0	157	24.6	0.017	70.7	214	25.9	0.056	1.64	cy	r + s
S5G-r-SWSC			35.6	164	29.8	0.018	73.6	218	28.2	0.045	1.4	cy	s
S6	1.86	22.2	29.7	146	—	0.020	82.0	196	—	0.085	—	f + cy	—
S7C-r-S			27.4	169	15.4	0.018	68.5	213	8.5	0.070	0.63	cy	s
S8C-sq-S			30.3	182	24.5	0.016	52.3	229	17	0.044	0.78	cy	s
S9C-r-SC			29.9	186	27.1	0.017	82.0	270	37.7	0.069	2.15	cy	s
S10C-r-R			31.6	160	9.6	0.015	73.6	251	28.1	0.055	1.53	cy	s
S11C-r-T			26.7	156	6.6	0.022	74.3	258	31.8	0.059	1.51	cy	s
S12C-st-T			26.8	174	18.9	0.019	74.5	249	27.1	0.042	1.43	cy	s
S13C-st-T-p/s			35.3	177	21.2	0.030	79.7	257	31.3	0.064	1.68 ^e	cy	s

^aInitial cracking only applies to control specimens (S1, S6) because all other specimens were precracked.^bCalculated for load with respect to control specimen (S1 for the first batch, and S6 for the second batch).^cNot applicable for control specimens. S1 and S6 are used in calculating the percentage increase of load in the strengthened specimens.^dFailure mode: f = flexure failure in concrete; cy = concrete crushing after steel yielding; r = rupture of FRP NSMR; s = splitting of epoxy cover.^eIncludes the initial strain caused by the prestressing.**Fig. 5.** Load-deflection response caused by material type effect**Fig. 6.** Load-deflection response caused by cross-sectional shape effect

response of all specimens. Compared with the corresponding control specimens (S1 and S6), the load increase at failure was approximately 8.5, 25.9, 28.1, 31.8, and 37.7% for specimens strengthened with smooth (S7C-r-S), SWSC (S4C-r-SWSC), ribbed (S10C-r-R), textured (S11C-r-T), and sand-coated-only (S9C-r-SC) rods, respectively. It can be seen that, except for the smooth surface, all other surface treatments can significantly increase the yield and load-carrying capacity of the concrete slab. This percentage increase is quite close for the ribbed, textured, and sand-coated-only conditions.

Effect of Prestressing

Specimen S13C-st-T-p/s was strengthened with three textured CFRP strips that were prestressed by using the method developed in this study. For a direct comparison, Specimen S12C-st-T used the same strips without any prestress. For some unknown reason, the two unstrengthened concrete slabs displayed a slightly different initial stiffness during their precracking test, causing different initial stiffness in the strengthened specimens (Fig. 8). Nevertheless, when compared with the control specimen (S6), both these two strengthened specimens displayed a significant increase in the load-carrying capacity, i.e., 27.1 and 31.3% for S12C-st-T and S13C-st-T-p/s, respectively. The tracking results on crack propagation in concrete during the test showed that the crack on the top and side surfaces of the prestressed specimen was developed much more slowly and the crack width was smaller than the nonprestressed specimen, indicating that the prestressed NSMR was effective in delaying the crack formation in concrete. Fig. 8 illustrates the load versus displacement response of the specimens, where the prestressed specimen (S13C-st-T-p/s) showed a greater stiffness

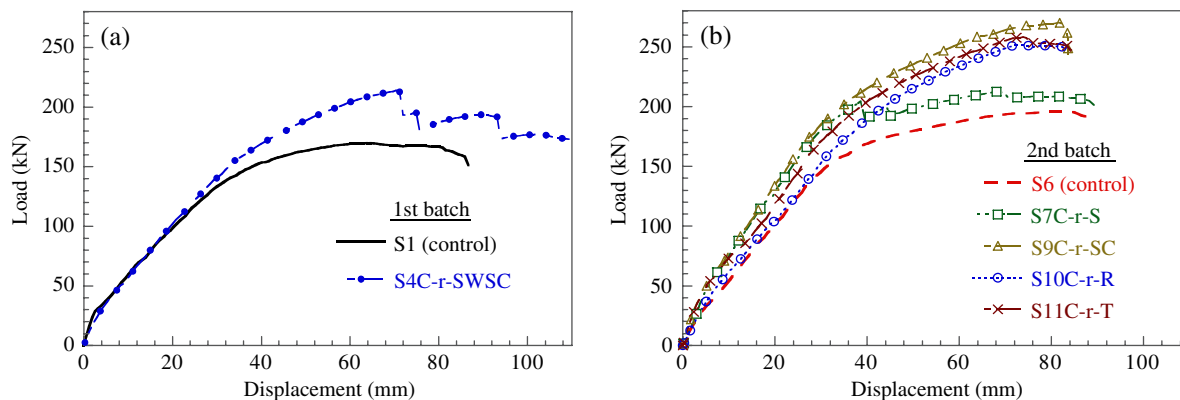


Fig. 7. Load-deflection response caused by surface treatment effect

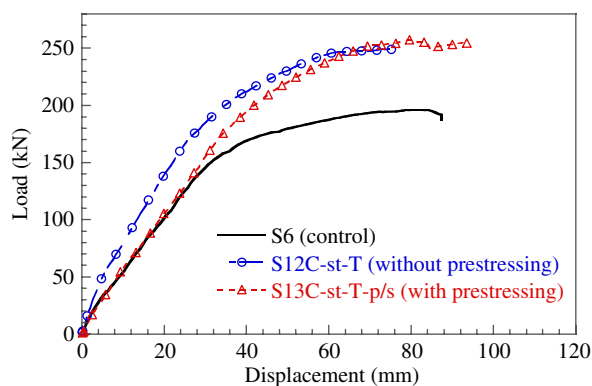


Fig. 8. Load-deflection response caused by prestressing effect

after the steel yielded and also a slightly greater load-carrying capacity than the nonprestressed specimen (S12C-st-T). However, the ductility was found to be less in S13C-st-T-p/s than that in S12C-st-T (as indicated in Table 3). Both specimens failed because of concrete crushing and epoxy splitting in the grooves. The maximum strain in the CFRP strips was found to reach 14,300 and 16,800 $\mu\epsilon$ at failure for S12C-st-T and S13C-st-T-p/s (this strain includes the initial strain caused by prestressing), respectively. The prestress applied here allowed for more efficient use of the NSM material than the nonprestressed case, resulting in an increase in the ultimate load. The strengthening effect caused by greater prestressing levels needs further investigation.

Analytical Predictions

An analytical study was conducted on the flexural behavior of both the unstrengthened and strengthened specimens using the

typical moment-curvature analysis method. The basic assumptions adopted here include (1) plane sections remain plane; (2) perfect bond between FRP NSM reinforcement and concrete; (3) the tensile strength of concrete is neglected; (4) the compressive behavior of concrete follows a parabolic stress-strain relationship proposed by Hognestad et al. (1955); (5) elastic-perfectly plastic behavior for steel reinforcement and linear-elastic behavior for FRP reinforcement; and (6) the failure mode is governed by the compressive failure in concrete. For a typical cross section of an NSM-strengthened concrete slab overhang, the internal stress and strain distribution along the cross section is represented in Fig. 9. The location of the neutral axis (c_u) is determined from the typical internal force equilibrium along the entire cross section, and the resulting compressive force in concrete (C_c) equals the total tensile force from the bottom and top layer of the steel reinforcement (T_{s1} and T_{s2}) and the FRP NSM reinforcement considering the preexisting stress caused by prestress (T_f). The moment capacity of the section (M_u) can then be calculated by summing the total contribution from each component. The curvature at ultimate (χ_u) is defined as the ratio between the ultimate compressive strain in concrete (ϵ_{cu}) and the distance of the neutral axis, i.e., $\chi_u = \epsilon_{cu}/c_u$. The analytical results thus obtained on total load, flexural moment capacity, and curvature were further compared with the experimental data obtained from the test, and a reasonable agreement was obtained, as seen in Table 4.

Ductility Requirement

As discussed in the previous section, the NSMR technique was found to be effective in increasing both the yield and load-carrying capacity of concrete overhanging slabs. To ensure the development of the plastic deformation in the steel reinforcement for desired warning prior to final failure in the strengthened concrete members, a minimum curvature ductility (defined by $\mu_\chi = \chi_u/\chi_y$) is suggested by the Federation Internationale du Beton (FIB) recommendation (FIB 2001), which proposes a minimum index of 1.7 for

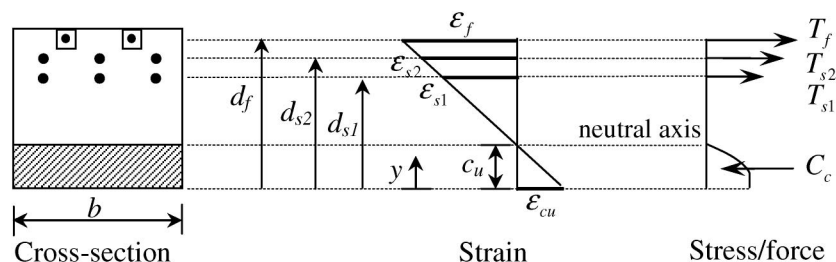


Fig. 9. Internal strain and stress distribution along cross section of strengthened specimens

Table 4. Comparison of Analytical Predictions and Experimental Results at Ultimate

Specimen designation	Load			Moment			Curvature		
	$P_{u(exp.)}^a$ (kN)	$P_{u(ana.)}^b$ (kN)	$P_{u(exp.)}/P_{u(ana.)}$	$M_{u(exp.)}$ (kN · m)	$M_{u(ana.)}$ (kN · m)	$M_{u(exp.)}/M_{u(ana.)}$	$\chi_{u(exp.)}$ (m ⁻¹)	$\chi_{u(ana.)}$ (m ⁻¹)	$\chi_{u(exp.)}/\chi_{u(ana.)}$
S1	170	174.7	0.97	222.8	228.9	0.97	0.031	0.036	0.85
S4C-r-SWSC	214	267.8	0.80	280.5	350.9	0.80	0.056	0.056	1.00
S5G-r-SWSC	218	234.9	0.93	285.7	307.8	0.93	0.045	0.063	0.71
S6	196	175.5	1.12	256.9	230.0	1.12	0.085	0.066	1.29
S7C-r-S	213	216.0	0.99	279.2	283.1	0.99	0.070	0.067	1.04
S8C-sq-S	229	223.1	1.03	300.1	292.3	1.03	0.044	0.065	0.68
S9C-r-SC	270	298.7	0.90	353.9	391.5	0.90	0.069	0.051	1.34
S10C-r-R	251	255.7	0.98	329.0	335.1	0.98	0.055	0.059	0.94
S11C-r-T	258	243.8	1.06	338.1	319.5	1.06	0.059	0.061	0.97
S12C-st-T	249	237.5	1.05	326.3	311.3	1.05	0.042	0.062	0.67
S13C-st-T-p/s	257	255.9	1.00	336.8	335.4	1.00	0.064	0.058	1.09

^aexp. = test result from the experiment.^bana. = predicted value from the flexural analysis.

steel S500 and concrete types C35/45 or lower. χ_u and χ_y are the curvature at ultimate and yielding, respectively. On the basis of the test results obtained in this study (Table 3), this minimum curvature ductility requirement is met for all NSM-strengthened concrete specimens, with the highest for S9C-r-SC and the lowest for S13C-st-T-p/s.

Bond Stress

The average bond stress of the FRP NSM reinforcement can be estimated from the strain response measured from the strain gauges attached to the NSMR surface. The following equations can be easily derived following the similar force equilibrium principle as used by Bonaldo et al. (2008) on a representative unit between two adjacent strain gages. For NSM circular rods and rectangular rods or strips, the average bond stress (τ_{avg}) can be calculated as

$$\tau_{avg} = \frac{d_b E_f \Delta \varepsilon}{4l} \quad \text{for circular rods} \quad (1)$$

$$\tau_{avg} = \frac{a_b b_b E_f \Delta \varepsilon}{2(a_b + b_b)l} \quad \text{for rectangular rods or strips} \quad (2)$$

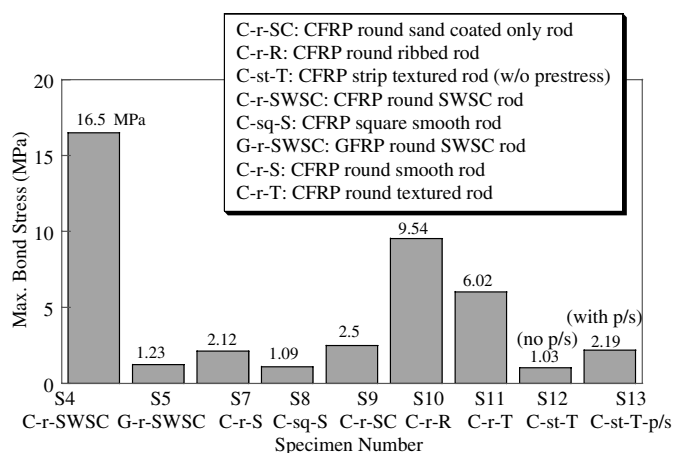
where τ_{avg} = average bond stress; d_b = diameter of the circular NSM rod; E_f = modulus of elasticity of the NSM FRP

reinforcement; $\Delta \varepsilon$ = axial strain difference between the two adjacent strain gauges on the FRP reinforcement; l = distance between the two adjacent strain gauges; and a_b, b_b = smaller and larger edge length of the rectangular NSM rod or strip. It was observed during the test that the bond stress of the NSM reinforcement tended to suddenly increase once the steel rebars yielded. The maximum bond stress developed in each type of FRP NSM reinforcement is summarized in Fig. 10, in which the SWSC CFRP round rod (S4C-r-SWSC) showed the highest bond stress of 16.5 MPa, and the nonprestressed textured CFRP strip (S12C-st-T) had the lowest bond stress of 1.03 MPa. The CFRP strip has the largest perimeter-to-sectional area ratio among all reinforcement types (Table 1). This bond stress was found to gradually increase from the free end to the support and increase as the load level was increased.

Conclusions

In this study, the strengthening effect of precracked bridge slab overhangs was investigated with regard to material type, cross-sectional shape, surface treatment, and use of prestress in the NSM reinforcements. Eleven full-scale reinforced concrete slab overhang specimens were tested, and an analytical study was performed. The following conclusions can be drawn:

1. The NSM technique was effective in increasing both the load at yielding of steel reinforcement (up to 27.1%) and at ultimate level (up to 37.7%) when compared with the unstrengthened RC members.
2. With the same amount of axial stiffness, the specimen strengthened by GFRP round rods showed similar flexural performance to the specimen with CFRP round rods (both had a spirally wound plus sand-coated surface treatment).
3. With a smooth surface, the square cross section was found to achieve a greater yield load in the specimen than the round cross section (slightly greater load-carrying capacity). This could be partially attributable to the wedging effect from the sharp edges of the square shape that increased the bond capacity.
4. The specimen strengthened with textured CFRP strips displayed performance close to the specimen using textured CFRP round rods, but the former achieved a greater yield load. The strip shape (rectangular) could be more efficient because of a larger perimeter-to-sectional area ratio than the round shape, which also reduced the bond stress.

**Fig. 10.** Maximum bond stress of all NSM reinforcements

5. In comparison with specimens strengthened with the same round cross-sectional shape of NSM reinforcement, all surface treatments of the CFRP NSMR except for the smooth condition significantly increased the yield and load-carrying capacity of the concrete slab.
6. The prestressing applied to the NSM strips was found to have a small improvement on the load-carrying capacity, but it was beneficial in delaying the formation of the crack in concrete. The prestressing method specially designed in this study was fairly simple and easy to implement in the laboratory.
7. The sectional analysis using the moment-curvature method was able to provide reasonable predictions on load, moment, and curvature for both strengthened and unstrengthened concrete slabs.
8. All the NSM-strengthened concrete slabs were able to meet the minimum curvature ductility requirement per FIB recommendation.

Several other issues remain to be investigated in the future research. A direct comparison of precracked (predamaged) and uncracked specimens strengthened by NSMR is needed because the existence of cracks affects the crack propagation and the debonding failure process of strengthened members. Additional analysis and direct pullout tests are desirable to further verify the wedging effect and the perimeter-to-sectional ratio in the NSM reinforcement. Moreover, the use of greater prestressing level and the field application of the developed prestressing unit need to be investigated in the future research.

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Notation

The following symbols are used in this paper:

- A_f = cross-sectional area of FRP NSM reinforcement (mm^2);
- a_b, b_b = smaller and larger edge length of a rectangular cross section for NSM rod or strip (mm);
- b = width of concrete slab section (m);
- C_c = compressive force in concrete (N);
- c_u = distance of neutral axis measured from bottom extreme fiber of the section at ultimate (mm);
- d_b = diameter of NSM reinforcement (mm);
- d_{s1}, d_{s2}, d_f = distance from extreme fiber to centroid of bottom layer, top layer of steel, and NSM reinforcement (mm);
- E_f = elastic modulus of FRP (MPa);
- l = distance between two consecutive strain gauges on NSM reinforcement (mm);
- M_u = ultimate flexural moment capacity of slab section (kN-m);
- P_{cr}, P_y, P_u = load at initial cracking, yielding and ultimate level (kN);

- T_{s1}, T_{s2}, T_f = tensile force in bottom layer, top layer of steel and NSM reinforcement (N);
- y = distance measured from bottom extreme fiber (mm);
- $\Delta\epsilon$ = strain difference between two consecutive strain gauges on NSMR;
- ΔP = load difference between strengthened and control specimens (kN);
- δ_y, δ_u = deflection at yielding and ultimate (mm);
- ϵ_{cu} = ultimate compressive strain in concrete (mm/mm);
- ϵ_{frp} = strain level in FRP composite at ultimate (mm/mm);
- μ_Δ = deflection ductility index, defined by δ_u/δ_y (mm/mm);
- μ_χ = curvature ductility index, defined by χ_u/χ_y ($\text{m}^{-1}/\text{m}^{-1}$);
- τ_{avg} = average bond stress (MPa); and
- χ_y, χ_u = curvature at yielding and ultimate (m^{-1}).

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